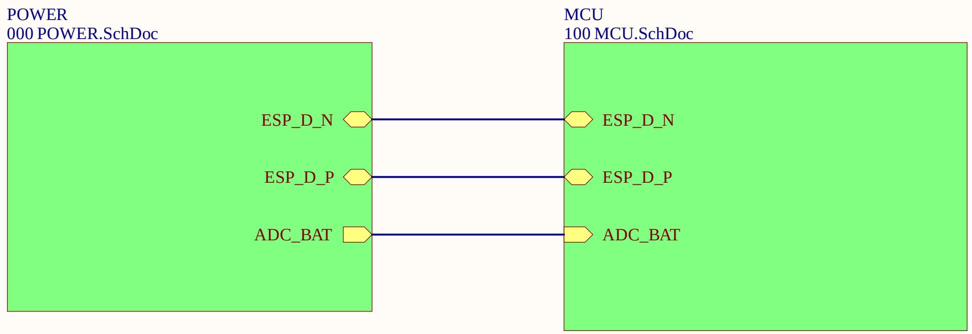

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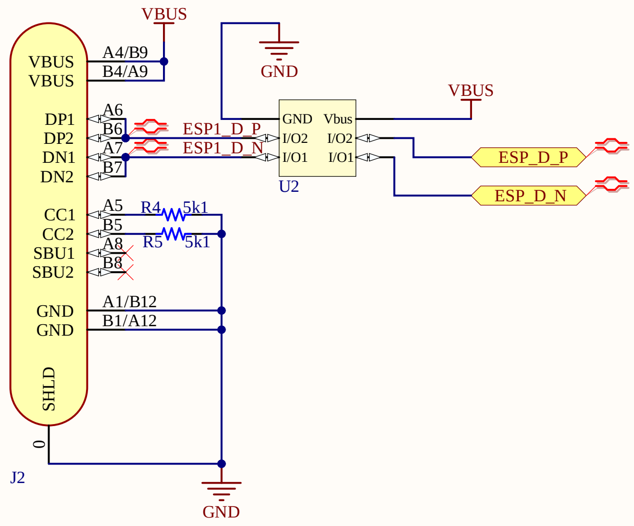
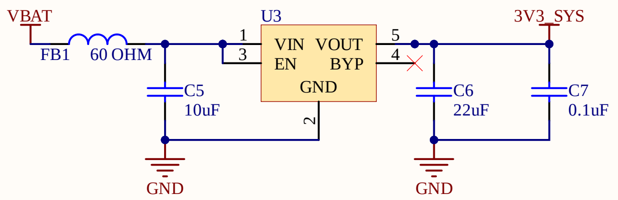
File:	C:\Diseño_electronico_Trabajos\..\Main.SchDoc	Drawn By:

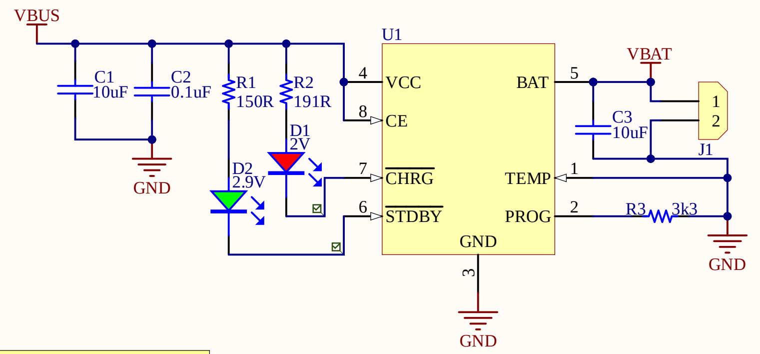
USB-C Power and Programming Interface



3.3 V System Regulator

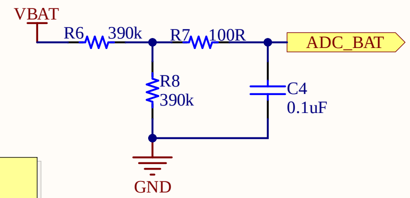


Li-Ion Battery Charger



RED LED:
Vf ≈ 2.1V, I ≈ 15.2 mA->R = 191R
GREEN LED:
Vf ≈ 2.9V, I ≈ 14 mA ->R = 150 R

Battery Voltage Monitor



Battery monitor:
R_TOP = 390k, R_BOTTOM = 390k
V_ADC = VBAT / 2
VBAT = 2 × V_ADC
I_DIV ≈ 5.4 µA @ VBAT = 4.2V
ADC_BAT connected to ESP32 IO0 / ADC1_CH0

Title		
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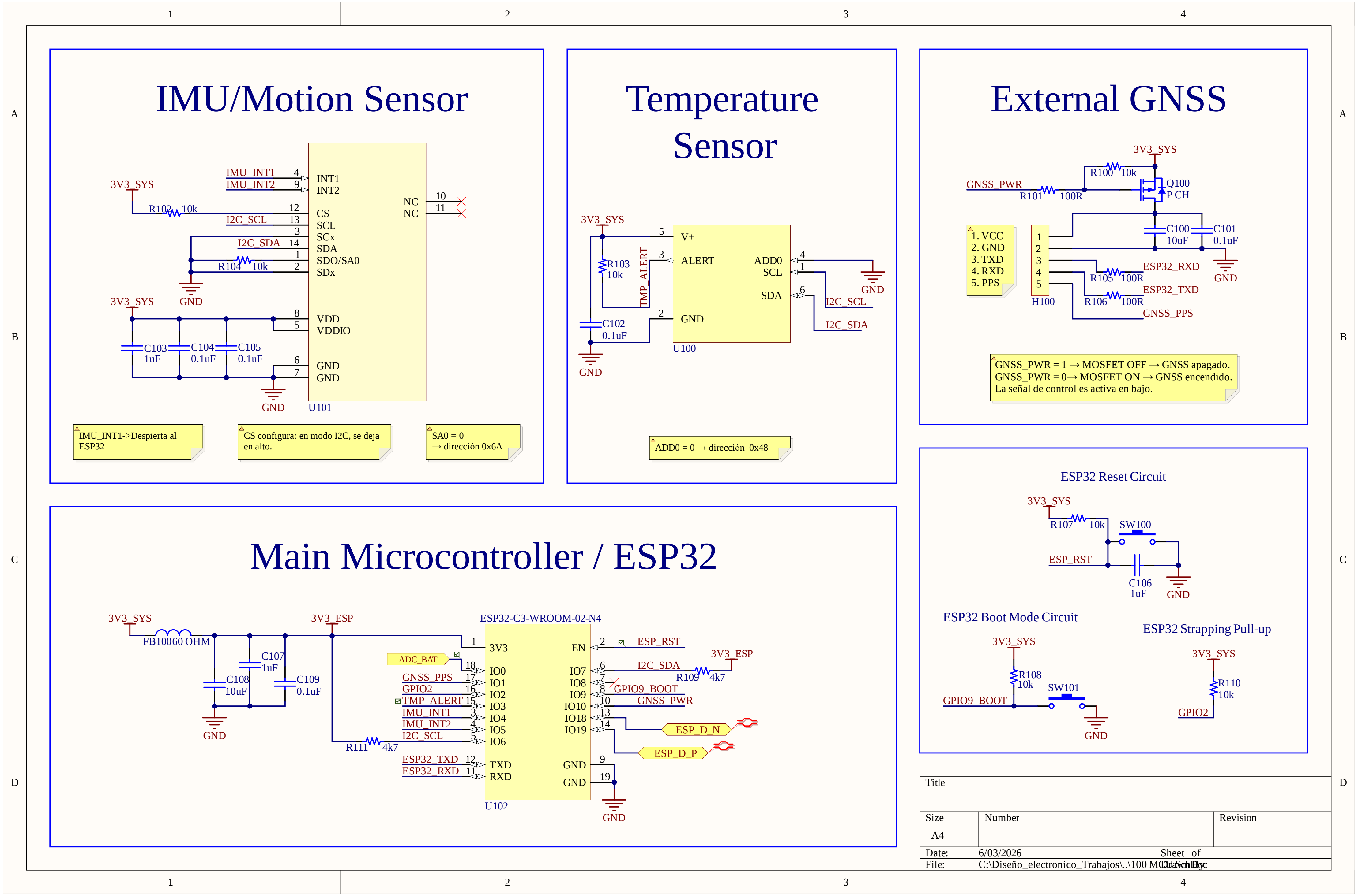

IMU/Motion Sensor
3V3_SYS
R102 10k
IMU_INT1 4 INT1
IMU_INT2 9 INT2
I2C_SCL 13 SCL
12 CS
3 SCx
I2C_SDA 14 SDA
1 SDO/SA0
R104 10k 2 SDx
NC 10
NC 11
GND
3V3_SYS
8 VDD
5 VDDIO
C103 1uF
C104 0.1uF
C105 0.1uF
6 GND
7 GND
GND
U101
IMU_INT1->Despierta al ESP32
CS configura: en modo I2C, se deja en alto.
SA0 = 0 → dirección 0x6A
Temperature Sensor
3V3_SYS
R103 10k
TMP_ALERT
5 V+
3 ALERT
ADD0 4
SCL 1
SDA 6
2 GND
C102 0.1uF
GND
I2C_SCL
I2C_SDA
U100
ADD0 = 0 → dirección 0x48
External GNSS
3V3_SYS
R100 10k
GNSS_PWR
R101 100R
Q100 P CH
C100 10uF
C101 0.1uF
GND
1. VCC
2. GND
3. TXD
4. RXD
5. PPS
H100
R105 100R ESP32_RXD
R106 100R ESP32_TXD
GNSS_PPS
GNSS_PWR = 1 → MOSFET OFF → GNSS apagado.
GNSS_PWR = 0→ MOSFET ON → GNSS encendido.
La señal de control es activa en bajo.
ESP32 Reset Circuit
3V3_SYS
R107 10k
SW100
ESP_RST
C106 1uF
GND
ESP32 Boot Mode Circuit
3V3_SYS
R108 10k
SW101
GPIO9_BOOT
GND
ESP32 Strapping Pull-up
3V3_SYS
R110 10k
GPIO2
Main Microcontroller / ESP32
3V3_SYS
FB10060 OHM
3V3_ESP
C107 1uF
C108 10uF
C109 0.1uF
GND
R111 4k7
ESP32-C3-WROOM-02-N4
1 3V3
EN 2 ESP_RST
ADC_BAT 18 IO0
GNSS_PPS 17 IO1
GPIO2 16 IO2
TMP_ALERT 15 IO3
IMU_INT1 3 IO4
IMU_INT2 4 IO5
I2C_SCL 5 IO6
ESP32_TXD 12 TXD
ESP32_RXD 11 RXD
IO7 6 I2C_SDA
R109 4k7
3V3_ESP
IO8 7
IO9 8 GPIO9_BOOT
IO10 10 GNSS_PWR
IO18 13 ESP_D_N
IO19 14 ESP_D_P
GND 9
GND 19
GND
U102
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Size A4
Number
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Date: 6/03/2026
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File: C:\Diseño_electronico_Trabajos\..\100 MCU.SchDoc
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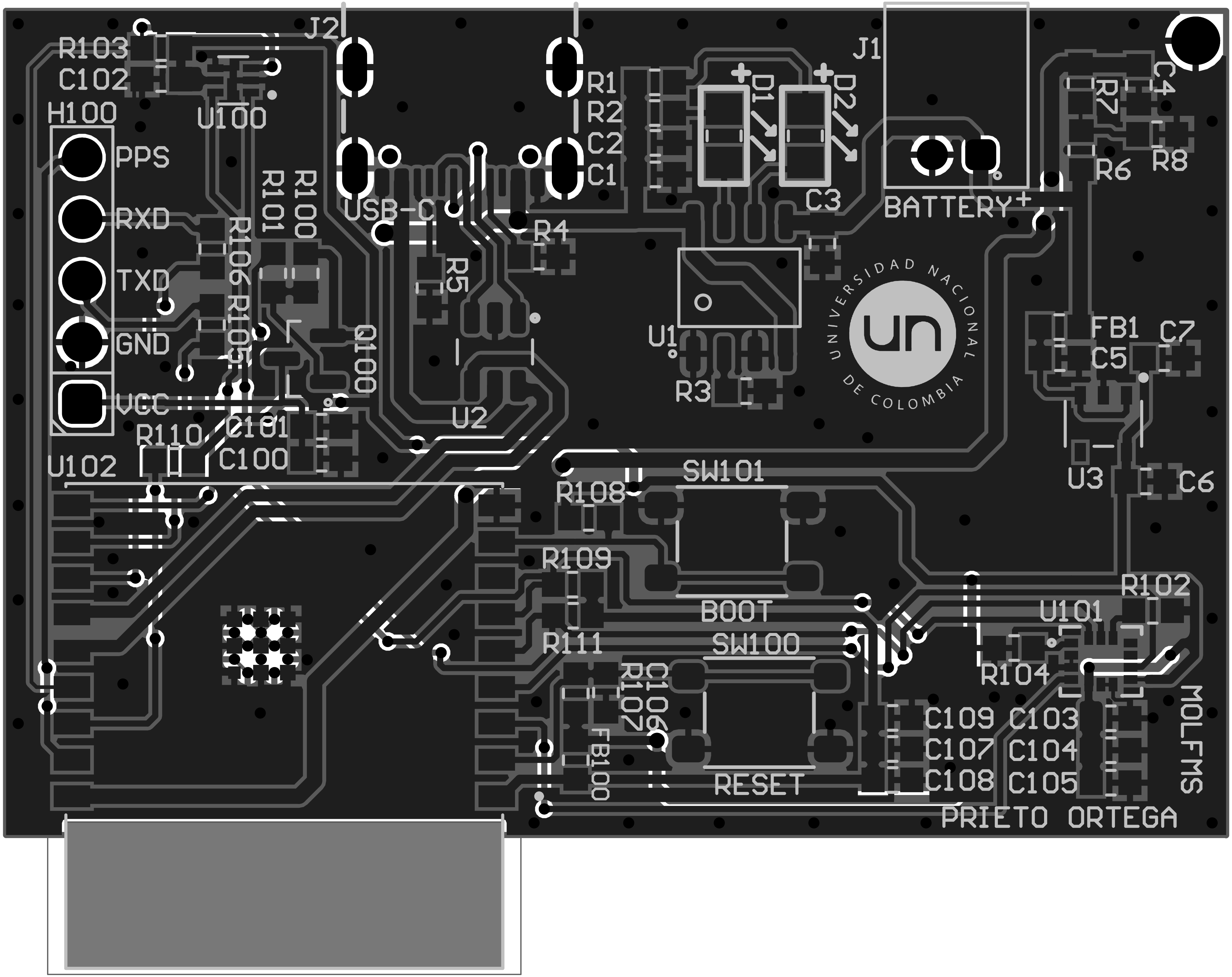

R103
C102
J2
U100
H100
PPS
RXD
TXD
GND
VCC
R101
R100
R106
R105
Q100
USB-C
R4
R5
U2
R1
R2
C2
C1
D1
D2
J1
BATTERY+
C3
U1
R3
UNIVERSIDAD NACIONAL DE COLOMBIA
C4
R7
R6
R8
FB1
C7
C5
U3
C6
R110
C101
C100
U102
R108
SW101
R109
BOOT
R111
SW100
R102
U101
R104
C106
R107
FB100
RESET
C109
C103
C107
C104
C108
C105
MOLFMS
PRIETO ORTEGA

Board Stack Report